

BHUMI

AI Land Intelligence Platform for India

Full Product Specification & Real-World Implementation Guide

Pan-India Land Records · AI Virtual Vakil · Real Lawyer Marketplace · 3D Infrastructure Modelling · Social Justice Alerts

Document Classification

Version: 1.0 | Status: Confidential – Internal Planning Document
Coverage: Complete product spec, datasets, tech stack, API sources, MVP scope, team, revenue model
Intended for: Founders, lead developers, investors, legal advisors

Module	Status
Land Records Engine	Core – MVP Phase 1
AI Virtual Vakil	Core – MVP Phase 1
Talk to Lawyer Marketplace	MVP Phase 2
3D Infrastructure Modelling	MVP Phase 2
Idle Land Scanner	MVP Phase 2
Social Justice Alert (Twitter/X)	MVP Phase 3
PWA + Offline Mode	Core – MVP Phase 1
Multilingual NLP (22 languages)	Core – MVP Phase 1

1. Executive Summary

Bhumi is a Progressive Web App (PWA) that gives every Indian citizen transparent, AI-powered access to land records, legal guidance, and social accountability tools — in their own language, on any device, even offline.

India has approximately 66 million land disputes pending in courts. Land conflicts constitute nearly 25% of all Supreme Court cases and are the single largest driver of rural poverty. The government's DILRMP programme has digitised 98.5% of rural land records — but has built no usable citizen-facing interface on top of that data. The data exists. No product surfaces it intelligently.

The Core Problem Bhumi Solves

A farmer in Agra has no idea who owns the adjacent plot that is being fenced off.

A slum resident in Mumbai cannot find out if government land 200 metres away sits vacant.

A widow in Bihar cannot afford a lawyer but her ancestral land mutation was fraudulently changed.

An architect in Chennai has no real parcel data to design on — they guess boundaries.

A journalist in Hyderabad cannot prove that a politician owns land through a proxy.

Bhumi solves all five of these. At once. For free.

1.1 What Makes Bhumi Unique

No existing platform in India combines:

- Full pan-India land parcel history (ownership chain, mutation dates, encumbrances, court status)
- An AI legal assistant that knows your specific parcel before you say a word
- A verified land revenue lawyer marketplace where every lawyer receives full context before the call
- 3D infrastructure planning on real GIS parcel geometry
- An idle land scanner that detects underutilised government or private land near slum zones
- Automatic social justice alerts posted to X/Twitter with evidence when exploitation is detected
- All of this in 22 Indian languages, offline-capable, free for citizens

2. Full Feature Specification

2.1 Land Records & History Engine

The core engine of the platform. Every land parcel in India has a unique 14-digit ULPIN (Unique Land Parcel Identification Number) assigned by DILRMP based on geo-coordinates. Bhumi uses ULPIN as the primary key for all records.

What It Shows Per Parcel

- Complete ownership chain from earliest available record to present day
- Every mutation entry: date, type (sale, inheritance, court order, govt acquisition), parties involved
- Encumbrance status: mortgages, liens, court attachment orders
- Survey map: cadastral boundary overlaid on satellite imagery
- Adjacent parcel ownership – who owns what surrounds your land
- Government land classification: revenue, forest, defence, urban local body, panchayat
- Current dispute status: whether any court case is registered against this parcel (eCourts API)
- Possession vs title split: shows cases where title holder and physical occupant differ
- Valuation history: stamp duty rates and registered transaction values

User Experience

User searches by: ULPIN, survey number, owner name, village/tehsil/district, or by dropping a pin on the map. The result is a timeline view — every event in the parcel's history shown chronologically with source citations. Every data point links back to the originating government record.

2.2 AI Virtual Vakil

The AI Virtual Vakil is a conversational legal assistant trained on Indian land and revenue law. It is always free, always available, and speaks any of 22 Indian languages. It is not a generic chatbot — it has read your specific land file before the conversation begins.

How It Works — Step by Step

1. User selects or searches their parcel. The AI auto-loads ownership chain, mutation history, court status, encumbrances, and survey data.
2. AI greets the user in their preferred language and presents a plain-language summary of the parcel's current legal standing.
3. AI asks a maximum of 3 intake questions: What has happened? When did it start? Who is the opposing party?
4. AI classifies the dispute into one of 8 categories (see below) and maps it to the applicable laws.
5. AI assesses claim strength: Strong / Moderate / Weak / Unclear, with reasoning.

6. AI recommends a next action: send legal notice, file RTI, approach revenue court, file police complaint, seek mediation, or escalate to human lawyer.
7. AI drafts the required document on request – RTI application, legal notice, mutation objection letter, complaint to DM.
8. If the case is complex, AI prepares a structured handoff brief for a real lawyer.

8 Dispute Categories & Applicable Laws

Dispute Type	Primary Laws Applied
Encroachment	Transfer of Property Act 1882 s.52, Specific Relief Act 1963 s.6, CPC Order 39
Mutation fraud	State Land Revenue Codes, Indian Penal Code s.420/467/468, Registration Act 1908
Inheritance / ancestral property	Hindu Succession Act 1956, Indian Succession Act 1925, Muslim Personal Law
Illegal possession / adverse possession	Limitation Act 1963 s.65, Specific Relief Act s.5-6
Government acquisition dispute	Land Acquisition Act 2013, LARR Act, Article 300A Constitution
Tenancy dispute	State Tenancy Acts (varies by state), Transfer of Property Act s.105-117
Title fraud / forged documents	IPC s.467-471, Registration Act 1908, IT Act 2000 s.66C
Boundary dispute	Survey & Settlement Acts, CPC s.26, Survey of India Standards

Languages Supported

Hindi, English, Bengali, Telugu, Marathi, Tamil, Gujarati, Urdu, Kannada, Odia, Malayalam, Punjabi, Assamese, Maithili, Sanskrit, Konkani, Sindhi, Dogri, Manipuri, Bodo, Santhali, Kashmiri — all 22 languages listed in the Eighth Schedule of the Indian Constitution. Voice input supported for all 22.

2.3 Talk to a Land Revenue Lawyer

A curated marketplace of verified land revenue lawyers across India. Unlike LawRato or MyAdvo — which are cold directories — Bhumi's marketplace is context-first: every lawyer receives the user's full land file, AI case summary, and dispute classification before the session begins.

Lawyer Profiles Include

- Bar Council registration number (verified against state Bar Council APIs)
- Specialisation tags: revenue court, civil court, High Court, Supreme Court
- State and district coverage
- Languages spoken

- Success rate on settled/won cases (self-reported, community-verified)
- Consultation fee (set by lawyer – platform takes 12% commission)
- Verified reviews from past clients
- Typical response time

Consultation Flow

9. User selects parcel → AI generates case brief (2-page summary of land history, dispute type, claim strength, relevant laws, recommended next steps)
10. User searches lawyers: filter by state, language, court type, fee range, rating
11. User books: instant (if lawyer online) or scheduled slot
12. Lawyer receives case brief 30 minutes before session – arrives prepared
13. Session via in-app video/voice/chat (WebRTC)
14. Post-session: AI transcribes the call, extracts action items, updates the living case file
15. If lawyer recommends filing: RTI or complaint drafted in-app, filed directly where possible

Revenue Model for Lawyer Feature

Model	Details
Platform commission	12% of each consultation fee – taken at payment
Lawyer subscription	Rs. 499/month for premium listing (higher placement, analytics)
Document drafting fee	Rs. 99 per document (RTI, legal notice) – free for BPL users
Citizens always free	No charge to users for search, AI advice, case file creation

2.4 3D Infrastructure Modelling for Architects

Architects and urban planners can import any land parcel into a 3D modelling workspace. The geometry is real — pulled from Survey of India cadastral maps and GIS data — not guessed. The workspace overlays FSI rules, setback requirements, and zoning regulations for that specific plot.

Features

- Parcel boundary imported as an accurate 2D footprint in 3D space
- Terrain elevation data from ISRO Bhuvan DEM (digital elevation model)
- Adjacent building heights from satellite/LiDAR where available
- Auto-applied FSI limits and building height restrictions per municipal zone
- Shadow analysis tool: simulates sunlight at any hour of any day
- Export to DXF, IFC, and PDF for use in AutoCAD, Revit, ArchiCAD
- Collaboration: multiple architects can work on the same model
- Overlay: proposed design vs satellite imagery of current state

Revenue Model for Architecture Feature

Rs. 2,999/month per firm seat. First 30 days free. Individual architects Rs. 499/month.

2.5 Idle Land Scanner

The most disruptive feature on the platform. The scanner cross-references four datasets to identify government or privately-owned land that is vacant, underutilised, or sitting idle — near areas where slum displacement or housing shortage is documented.

How the Scanner Works

16. Platform ingests slum boundary data from MoHUA (Ministry of Housing & Urban Affairs) and state slum rehabilitation authorities.
17. Platform ingests government land classification data: plots owned by Municipal Corporation, State Government, Central Government, Railways, Defence, port trusts.
18. Satellite imagery (Sentinel-2, Bhuvan) is analysed via computer vision to classify land use: built-up, agricultural, vacant, water body, forest.
19. Scanner flags: any government-owned parcel within 2km of a documented slum zone that satellite analysis shows as >70% vacant/undeveloped.
20. Scanner also flags: private land parcels >1 acre, no development for 5+ years, registered to individuals or shell companies, within the same proximity.
21. Flagged parcels are ranked by: size, proximity to slum, owner category, years idle.
22. Result is displayed as a heatmap overlay on the city map.

Why This Matters

Under the Urban Land (Ceiling and Regulation) Act (now repealed in most states but with residual provisions) and the Smart Cities Mission guidelines, governments are required to prioritise idle public land for affordable housing. The scanner creates public evidence that this is not happening — which feeds directly into the Social Justice Alert system.

2.6 Social Justice Alert System

When the Idle Land Scanner detects a case — a slum is being demolished or displaced while government land sits idle nearby — the platform automatically compiles an evidence package and posts a structured thread to X (Twitter) and optionally to Instagram and Facebook.

The Evidence Package Contains

- Satellite image of the idle government land (before/after comparison if displacement has begun)
- Land record showing current government owner and years since last mutation
- Slum boundary map showing the affected community
- Distance between idle land and slum zone
- Estimated number of families affected
- Relevant government scheme (PMAY, RAY) that should have applied
- Names of relevant municipal commissioner, DM, and elected representative — tagged in tweet

The Tweet Thread Structure

Tweet 1: 'ALERT: [N] families in [Slum Name], [City] face eviction. [X] acres of [Govt Body] land sits idle [Y] km away.'

Tweet 2: Satellite image comparison – idle land vs slum area.

Tweet 3: Land record screenshot. Owner: [Govt Body]. Last mutation: [Year]. Classification: [Urban/Revenue].

Tweet 4: 'Under PMAY guidelines, this land should have been allocated for EWS housing. DM @[handle] Municipal Commissioner @[handle] — what is the plan?'

Tweet 5: Link to full evidence report on the Bhumi platform, open for public verification.

User-Triggered vs Auto-Triggered

Any citizen or journalist can also manually trigger an alert for any parcel — the platform will generate the evidence package from available data. Manual alerts go through a 24-hour community moderation queue before posting. Automated scanner alerts for flagged cases post immediately.

2.7 PWA + Offline Capability

Bhumi is built as a Progressive Web App — installable on any Android or iOS device, no app store required. It works offline with locally cached land records for parcels the user has previously viewed. This is critical for rural and semi-urban users with unreliable connectivity.

Offline Capabilities

- View previously loaded parcel history and ownership data
- Continue AI Vakil conversation (cached context – AI response queued for sync)
- View downloaded survey maps
- Draft RTI applications and legal notices offline – submitted when connection returns
- 3D models cached locally after first load

Languages and Accessibility

- Language auto-detected from device locale on first launch
- Voice input for all 22 scheduled languages (Web Speech API + custom model for Indian accents)
- Screen reader compatible (WCAG 2.1 AA)
- Low-bandwidth mode: text-only interface for 2G connections

3. Datasets — Complete List with Sources

This section documents every dataset required to build and operate Bhumi, including source, access method, update frequency, format, and any known gaps.

3.1 Land Records Datasets

Dataset	Source	Access	Format	Frequency
ULPIN-tagged parcel records	DILRMP / DoLR	State NIC APIs + data.gov.in	JSON/XML	Daily sync
Bhulekh / ROR (Record of Rights)	State Revenue Depts (28 states)	Each state portal API or scrape	HTML/JSON	Daily
Mutation register (Fard)	Tehsil Revenue Office	State APIs: UP Bhulekh, Maharashtra 7/12, Karnataka Bhoomi	JSON	Daily
Encumbrance Certificate data	Sub-Registrar offices	CERSAI for mortgages; state EC portals	PDF/JSON	Real-time
Property registration (deed data)	IGRS state portals	State IGRS APIs (Andhra, UP, MH, TN)	JSON	Daily
Jamabandi / Pahani records	Punjab, Haryana, AP state portals	Direct API access	JSON	Daily
Urban land records (ULB)	Municipal Corporation GIS cells	City-specific APIs or CPGRAMS portal	Shapefile/JSON	Monthly
SVAMITVA scheme data (rural)	Ministry of Panchayati Raj	svamitva.nic.in API	JSON/GeoJSON	Weekly

3.2 GIS & Cadastral Map Datasets

Dataset	Source	Access	Notes
Cadastral maps (village-level)	Survey of India	NSDI portal (nsdi.gov.in) – licensed	1:3960 scale. State survey depts hold originals.
ISRO Bhuvan satellite imagery	ISRO / NRSC	bhuvan.nrsc.gov.in API (free for govt use)	2.5m resolution. Updated quarterly.

Dataset	Source	Access	Notes
Sentinel-2 satellite imagery	ESA Copernicus	Open access via AWS/Google Earth Engine	10m resolution. Free. 5-day revisit cycle.
Digital Elevation Model (DEM)	ISRO Bhuvan Cartosat	bhuvan.nrsc.gov.in	30m resolution. Used for 3D modelling terrain.
Urban cadastral (city-level)	NUIS (National Urban Info System)	MoHUA / Town Planning Dept	Varies by city. Chennai, Delhi most complete.
Forest land boundary	FSI (Forest Survey of India)	fsi.nic.in	Critical for encroachment detection.
Railway land boundary	Railways land records	IRCTC / Ministry of Railways (RTI)	Often not digitised – gap dataset.
Defence land boundary	Military Engineer Services	Classified – use buffer zone approach	Use proximity exclusion in scanner.
Slum boundary maps	MoHUA / State SRAs	mohua.gov.in + state portal data	Updated post-census. 2011 data available.

3.3 Legal & Court Datasets

Dataset	Source	Access	Notes
Court case status per property	eCourts API	api.ecourts.gov.in (open API)	Search by survey number or party name. National High Court + District Court data.
Supreme Court orders	Supreme Court of India	sci.gov.in / NJDG	For land acquisition and constitutional cases.
Revenue court orders	State Revenue Tribunals	State-specific portals (e.g. UP Board of Revenue)	Partially digitised. Many require RTI.
RERA registered projects	State RERA portals	Each state RERA API	Builder-buyer dispute data. MH RERA most complete.
Legal aid case data	NALSA / SLSA	nalsa.gov.in	Free legal aid eligibility and case tracking.
Lok Adalat outcomes	NALSA	nalsa.gov.in/lok-adalat	Settlement data for land disputes.
Indian Kanoon case law	Indian Kanoon (private)	indiankanoon.org API	10M+ judgments. Used for AI Vakil legal reasoning.
SCC Online / Manupatra	Subscription databases	Licensed API – paid	For precedent matching in AI. ~Rs.2L/year.

3.4 Government Land Classification Datasets

Dataset	Source	Access	Notes
Government land inventory	DARPG / DoLR	data.gov.in	Central govt land parcels. Incomplete — gaps filled by RTI.
State government land register	State Revenue Depts	State NIC portals	Varies by state. UP, MH, KA most digitised.
Urban local body (ULB) land	Municipal Corporation GIS	City-specific portals	Mumbai, Delhi, Bengaluru have GIS portals.
Railways land parcels	Ministry of Railways	RTI + partial digitisation	Significant gap — file RTIs systematically.
Port trust land	Ministry of Ports	portsadmin.nic.in	12 major ports. Boundary data available.
Land Acquisition notifications	State Gazette	State e-gazette portals	Critical for detecting displacement alerts.
PM Awaas Yojana allotments	MoHUA PMAY portal	pmaymis.gov.in API	Shows which land has been allocated for EWS housing.

3.5 AI Training & Knowledge Datasets

Dataset	Purpose	Source	Notes
Transfer of Property Act 1882	Legal reasoning	Legislative.gov.in (open)	Full text. All amendments.
Land Acquisition Act 2013 (LARR)	Legal reasoning	Legislative.gov.in	Critical for govt acquisition disputes.
Hindu Succession Act 1956	Inheritance disputes	Legislative.gov.in	Amended 2005 — daughters' rights.
All 28 State Land Revenue Codes	State-specific law	State govt websites	Must be collected state by state. Major gap.
All 28 State Tenancy Acts	Tenancy disputes	State portals	Highly variable — requires legal review per state.
10M+ Indian court judgments	Precedent matching	Indian Kanoon API	Land & property subset: ~2.3M judgments.
RTI templates — land disputes	Document drafting	RTI India, MyGov	500+ verified templates. Needs curation.
Mutation objection letters	Document drafting	Collected from NGOs / legal aid orgs	Multilingual templates required.
Legal notice templates (land)	Document drafting	Bar Council sample forms	State-specific formatting required.
Paralegal training corpus	AI Vakil dialogue	NALSA / Law Commission	For citizen-appropriate (non-jargon) responses.

3.6 Satellite & Urban Planning Datasets

Dataset	Purpose	Source	Notes
Sentinel-2 multispectral	Idle land detection	ESA Copernicus (free)	NDVI analysis for vegetation vs vacant land.
ISRO Bhuvan Cartosat	High-res urban mapping	bhuvan.nrsc.gov.in	0.65m resolution. Best for city-level analysis.
Google Earth Engine	Land use change detection	earthengine.google.com (free for research)	5-year change detection for idle land scanner.
OpenStreetMap	Base map layer	openstreetmap.org (open)	Road network, building footprints.
GHSL (Global Human Settlement Layer)	Slum detection	EU Joint Research Centre (open)	Urban/rural/slum classification at 10m.
WorldPop settlement data	Population density	worldpop.org (open)	Used to estimate affected families in alerts.
Census 2011 slum data	Slum boundary baseline	Registrar General of India	Most complete slum enumeration dataset.
Census 2031 (when released)	Updated slum baseline	Office of Registrar General	Monitor for release – will require integration.

4. Technology Stack

4.1 Frontend (PWA)

Layer	Technology	Reason
Framework	Next.js 14 (App Router)	SSR + SSG for SEO. PWA support via next-pwa. React ecosystem.
Language	TypeScript	Type safety critical for legal data accuracy.
UI components	Tailwind CSS + shadcn/ui	Rapid build. Accessible by default.
Maps	Maplibre GL JS	Open-source. Works with self-hosted tiles. Offline support.
Tile server	PMTiles (Protomaps)	Single-file map tiles. Can be served from CDN. Offline-capable.
3D modelling	Three.js + IFC.js	Web-based 3D. IFC.js for BIM format support.
Offline storage	IndexedDB via Dexie.js	Land records cached locally. Sync on reconnect.
Voice input	Web Speech API + Whisper (fallback)	Browser-native for online. Whisper model for offline/accents.
PWA	Workbox (service worker)	Cache strategies per route. Background sync for offline actions.
State management	Zustand	Lightweight. Works well with offline-first patterns.
Forms	React Hook Form + Zod	Validated document drafting forms.
i18n	next-intl	22 language support. RTL for Urdu.

4.2 Backend

Layer	Technology	Reason
API framework	FastAPI (Python)	Async. Auto OpenAPI docs. Strong ML/AI library ecosystem.
Language	Python 3.12	Best AI/ML ecosystem. Matches team skills for land data processing.
Auth	Supabase Auth + JWT	Aadhaar-optional. Mobile OTP (SMS) primary. Email secondary.
Database – primary	PostgreSQL 16 + PostGIS	PostGIS for geospatial queries. Land boundary intersection.
Database – cache	Redis 7 (Upstash)	Session cache. Rate limiting. API response cache.

Layer	Technology	Reason
Database — documents	MongoDB Atlas	Court orders, RTI drafts, case files — semi-structured.
Search	Elasticsearch / OpenSearch	Full-text search across 22 languages. Land owner search.
File storage	AWS S3 / Cloudflare R2	Satellite images, survey maps, 3D model exports, PDFs.
Task queue	Celery + Redis	Async: idle land scan jobs, alert posting, PDF generation.
Data pipeline	Apache Airflow	Scheduled ingestion of 30+ govt data sources. DAG per state.
Geospatial processing	GDAL + Shapely + Fiona	Coordinate transforms, boundary intersections, GIS ops.

4.3 AI / ML Stack

Component	Technology	Notes
AI Vakil LLM	Claude API (Anthropic) / GPT-4o	Primary reasoning. Context window holds full land file + conversation.
Indian legal fine-tune	LLaMA 3 fine-tuned on Indian judgments	10M judgment corpus from Indian Kanoon. Self-hosted.
Multilingual NLP	IndicBERT / MuRIL	Google's multilingual model for Indian languages. Open source.
Voice STT	Whisper Large v3 (OpenAI)	Supports all 22 scheduled languages. Self-hosted on GPU.
Satellite image analysis	Segment Anything Model (Meta) + custom classifier	Idle land detection from satellite imagery.
Land use classification	Random Forest on Sentinel-2 bands	NDVI + NDBI indices for land cover classification.
OCR (old land records)	Tesseract + custom Devanagari model	For scanned mutation registers not yet digitised.
Document drafting	Claude API with legal templates	RTI, legal notices, mutation objections — template-grounded generation.
Embedding / semantic search	sentence-transformers (multilingual)	For finding similar cases in the judgment corpus.
Vector database	Pinecone / pgvector	Store legal judgment embeddings for AI Vakil precedent lookup.

4.4 Infrastructure

Component	Technology	Notes
Cloud provider	AWS (primary) + Cloudflare (edge)	AWS for compute/storage. Cloudflare Workers for edge caching.
Container orchestration	AWS ECS / Fargate	Serverless containers. Auto-scale for peak land record queries.
CDN	Cloudflare	Map tiles, satellite images, static assets. India edge POPs in Mumbai, Delhi, Chennai.
Video calls (lawyer feature)	Daily.co or 100ms (Indian)	WebRTC. Low-latency. Compliant with Indian data laws.
SMS (OTP)	MSG91 or Exotel	Indian SMS gateway. OTP for Aadhaar-optional auth.
Email	AWS SES	Transactional emails.
Monitoring	Grafana + Prometheus + Sentry	API performance, error tracking, uptime.
CI/CD	GitHub Actions	Test → build → deploy pipeline per environment.
Data residency	AWS ap-south-1 (Mumbai)	All Indian citizen data stored in India. PDPB compliance.

4.5 Key External APIs

API	Purpose	Cost / Access
eCourts API (api.ecourts.gov.in)	Court case status per property	Free. Govt API key required.
DILRMP state NIC APIs	Land record data	Free for approved integrations. MoU required with DoLR.
ISRO Bhuvan API	Satellite imagery + DEM	Free for Indian companies. Registration required.
Copernicus/Sentinel-2 via Google Earth Engine	Idle land satellite analysis	Free for non-commercial. Rs.0/month up to usage limits.
Indian Kanoon API	Court judgment search for AI Vakil	Paid. ~Rs.1.5L/year for full access.
X (Twitter) API v2	Social justice alert posting	Basic tier: \$100/month. 50,000 posts/month.
MSG91 SMS API	OTP authentication	~Rs.0.20 per OTP. Bulk pricing available.
Anthropic / Claude API	AI Vakil reasoning engine	Pay per token. ~Rs.5-15 per full case analysis.
Google Maps / MapTiler	Base map layer fallback	MapTiler: \$25/month for 100K tile requests.

API	Purpose	Cost / Access
Daily.co WebRTC	Lawyer video calls	\$0.0045/participant-minute. ~Rs.10 per 30-min call.
CERSAI API	Mortgage/encumbrance data	Rs.50 per query. Bulk MoU possible.
State IGRS portals	Property registration data	Each state has separate integration. Most have APIs.

5. Data Ingestion Pipeline

Bhumi's data pipeline must ingest, clean, standardise, and sync records from 30+ government sources across 28 states and 8 Union Territories. This is the hardest engineering problem in the platform — not the AI, not the frontend.

5.1 Pipeline Architecture

The pipeline runs on Apache Airflow. Each state has a dedicated DAG (Directed Acyclic Graph) with its own schedule, schema mapper, and error handling. All records are normalised to a common Bhumi data schema before writing to PostgreSQL.

Core Pipeline Steps (per state DAG)

23. Extract: pull from state API or scrape state portal (Playwright-based scraper where no API exists)
24. Validate: check for missing ULPIN, malformed survey numbers, invalid district codes
25. Transform: normalise to Bhumi schema (ULPIN as PK, standardised owner name, date formats, classification codes)
26. Deduplicate: match against existing records — detect updates vs new parcels
27. Enrich: join with GIS boundary from Survey of India cadastral dataset
28. Enrich: check eCourts API for any case registered against this survey number
29. Enrich: run OCR if source is a scanned PDF (pre-2010 mutation records)
30. Load: write to PostgreSQL + update Elasticsearch index
31. Audit: log every change with source URL, pull timestamp, and data hash for traceability

5.2 State Coverage — Priority Order

Priority	States	Data Quality	API Available
Phase 1 (MVP)	Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, Telangana	High — well digitised	Yes — NIC APIs + state portals
Phase 2	Gujarat, Rajasthan, Madhya Pradesh, Andhra Pradesh, West Bengal	Medium	Partial — mixed API + scrape
Phase 3	Bihar, Odisha, Jharkhand, Chhattisgarh, Punjab, Haryana	Medium	Partial
Phase 4	Kerala, Assam, Himachal Pradesh, Uttarakhand, Goa	Medium-Low	Limited — heavy scraping
Phase 5	NE states, J&K, Ladakh, Lakshadweep, A&N Islands	Low — least digitised	Mostly manual / RTI

5.3 Handling Data Gaps

Not all records are digitised. Bhumi handles gaps with three approaches:

- RTI filing bot: auto-files RTI applications to tehsil offices for missing records in target areas. Responses ingested when received.
- Citizen contribution: users can upload physical land documents (7/12, Jamabandi, old patta) — OCR + human review adds them to the database.
- Community verification: disputes between records are flagged for community verification — local land records officers or nearby owners can confirm.

6. MVP Scope & Build Timeline

6.1 MVP Phase 1 — Core (Months 1–6)

Phase 1 Goal

A working PWA that lets any citizen in 5 states search their land parcel, see its full history, and get AI legal guidance in Hindi and English. No lawyer marketplace. No 3D. No Twitter alerts. Just the core data + AI Vakil + offline PWA.

Feature	Month	Team
Data pipeline – UP, MH, KA, TN, TG	1-2	2 backend engineers
PostgreSQL schema + PostGIS setup	1	1 backend engineer
ULPIN-based parcel search API	2	1 backend engineer
Land history timeline UI	2-3	1 frontend engineer
GIS parcel map (Maplibre + PMTiles)	3	1 frontend engineer
AI Vakil — core dispute analysis (Hindi + English)	3-4	1 AI engineer
AI Vakil – 8 dispute type classification	4	1 AI engineer
RTI and legal notice document drafting	4-5	1 AI engineer
PWA + offline mode (Workbox + Dexie.js)	5	1 frontend engineer
OTP auth (Aadhaar-optional)	2	1 backend engineer
eCourts API integration (court status)	3	1 backend engineer
Beta launch – 5 states, Hindi + English	6	Full team

6.2 MVP Phase 2 — Growth (Months 7–12)

Feature	Month	Team
10 more Indian languages	7-8	1 AI + 1 frontend
Lawyer marketplace – listing + booking	7-9	1 backend + 1 frontend
WebRTC video/voice calls	9-10	1 backend engineer
Post-call AI transcription + case file update	10	1 AI engineer
3D modelling workspace (Three.js + IFC.js)	8-11	1 frontend engineer
Idle land scanner – 5 pilot cities	9-11	1 AI + 1 data engineer

Feature	Month	Team
Satellite analysis pipeline (Sentinel-2 + Earth Engine)	9-10	1 data engineer
Expand to 15 states	8-10	1 data engineer
Architect subscription billing	10	1 backend engineer

6.3 MVP Phase 3 — Impact (Months 13–18)

Feature	Month	Team
Social justice alert system (X/Twitter integration)	13-14	1 backend + 1 AI
Alert moderation queue + community verification	14	1 backend
Expand to all 28 states + 8 UTs	13-16	2 data engineers
All 22 Indian languages	14-15	1 AI engineer
Offline voice input (Whisper local model)	15	1 AI engineer
RTI filing bot (auto-file for gaps)	15-16	1 backend
Public evidence dashboard for journalists	16	1 frontend
Mobile app shell (Capacitor wrapper)	17	1 frontend

7. Team Required to Build This

Role	Count	Key Skills	Phase
Full-stack / backend lead	1	Python, FastAPI, PostgreSQL + PostGIS, Airflow, AWS	Phase 1
Data engineer	1-2	Python, ETL, GDAL, scraping, govt API integrations	Phase 1
Frontend engineer	1	Next.js, TypeScript, Maplibre, PWA/Workbox, Dexie.js	Phase 1
AI / ML engineer	1	LLM APIs, fine-tuning, RAG, multilingual NLP, vector DBs	Phase 1
GIS specialist	1	QGIS, GDAL, Shapely, satellite imagery, land survey data	Phase 2
Legal advisor / domain expert	1	Land revenue law, state-specific codes, RTI process	Phase 1 — critical
UI/UX designer	1	Mobile-first, multilingual UI, accessibility	Phase 1
DevOps	1 (part-time)	AWS ECS, Cloudflare, CI/CD, monitoring	Phase 1
Govt relations / BD	1	MoU with DoLR, state NIC APIs, Bar Council	Phase 1 — critical
Community / operations	1-2	Lawyer onboarding, citizen support, content moderation	Phase 2

Critical Hire Warning

The Legal Advisor and Government Relations roles are as important as any engineer. Without a proper MoU with the Department of Land Resources (DoLR), you cannot legally access or republish state land record APIs. Without a qualified land law advisor reviewing the AI Vakil's outputs, you face liability for wrong legal guidance. These are not optional.

8. Revenue Model

Bhumi is free for citizens. Revenue comes from professional and institutional users, not from people who need land justice.

Revenue Stream	Target User	Price	Est. Year 1 Revenue
Architect/planner subscriptions	Architecture firms	Rs. 2,999/seat/month	50 seats × Rs.36K = Rs.18L
Lawyer marketplace commission	Land lawyers	12% of consultation fee	1000 consults/month × Rs.500 avg × 12% = Rs.7.2L/yr
Premium lawyer listing	Lawyers	Rs. 499/month	200 lawyers × Rs.6K = Rs.12L/yr
Document drafting fee	General users (non-BPL)	Rs. 99 per document	5000 docs/month × Rs.99 = Rs.59L/yr
Journalist / NGO data API	Media, research orgs	Rs. 25,000/year	20 orgs = Rs.5L/yr
Government licensing	State govts	Rs. 50L+ per state MoU	1 state pilot = Rs.50L
CSR grants	Corporates (Tata, HDFC, etc.)	Project-based	Rs.25-100L grants

Year 1 conservative estimate (Phase 1 live + early Phase 2): Rs. 1.5-2 Crore ARR. Year 3 target (all states live): Rs. 15-25 Crore ARR.

9. Legal, Compliance & Risk

9.1 Data Licensing

- All government data used must be attributed and used under the National Data Sharing and Accessibility Policy (NDSAP 2012) or state-equivalent.
- A Memorandum of Understanding (MoU) with the Department of Land Resources is required before bulk access to DILRMP data.
- State-level data sharing agreements required with each state NIC / Revenue Department — budget 6–12 months for approvals.
- Commercial use of court data (eCourts) requires a separate agreement. Non-commercial API use is open.

9.2 AI Legal Advice Liability

- The AI Vakil must display a clear disclaimer: 'This is legal information, not legal advice. Consult a qualified lawyer before taking any legal action.'
- Document drafts generated by AI must be reviewed by the user before submission.
- All AI legal outputs must be reviewed quarterly by a qualified advocate for accuracy drift.
- Platform is not liable for outcomes from AI-drafted documents used without lawyer review — this must be in the Terms of Service.

9.3 Data Privacy

- Personal Digital Data Protection Bill (PDPB) 2023 compliance mandatory. User data stored in India (AWS Mumbai).
- Aadhaar integration requires UIDAI sandbox access + DPDP consent framework — defer to Phase 2.
- All citizen case files are private by default. Land record data is public (already in government portals).
- Social justice alerts must not identify individual slum residents by name — use aggregate counts only.

9.4 Twitter/X Alert Legal Risk

- All automated posts must be factually sourced from government records — no editorial claims.
- Tag government officials only (not private individuals) in alerts.
- Maintain a 24-hour moderation window for citizen-triggered alerts (not auto-alerts from scanner).
- Retain legal counsel to respond to defamation claims. Structure alerts as 'questions' not 'accusations.'

10. Known Gaps, Risks & Mitigations

Risk	Severity	Mitigation
Govt API access denied / delayed	High	File RTIs in parallel. Build scrapers as fallback. Start with open data first.
State data in incompatible formats	High	Build a per-state schema mapper. Allocate 40% of data eng time to this.
AI Vakil gives wrong legal advice	High	Mandatory disclaimer. Quarterly legal audit. Scope to 'guidance' not 'advice.'
X/Twitter API becomes unaffordable	Medium	Also post to Mastodon, Bluesky. Build email alert list as primary channel.
Lawyer marketplace lawyer quality	Medium	Bar Council verification. Client review system. Remove poorly rated lawyers.
Pre-2000 land records not digitised	Medium	OCR pipeline + citizen document upload. RTI bot for missing tehsil records.
Satellite data latency for scanner	Low	Sentinel-2 is 5-day revisit. Accept 1-week lag on idle land detection.
Political pressure to suppress alerts	High	Data is public govt records – legally defensible. Have legal team ready.
Scale: 1B+ parcels to process	Medium	Shard PostgreSQL by state. PostGIS optimised indexes. Cache hot parcels in Redis.
Language model quality for dialects	Medium	Use MuRIL + IndicBERT ensemble. Add dialect-specific fine-tuning in Phase 3.

11. Competitive Moat

Bhumi's defensibility comes from four compounding advantages that get stronger the longer the platform operates:

1. Data Network Effects

Every citizen who uploads a missing document, confirms a boundary, or flags a dispute error improves the database for everyone. After 2 years, Bhumi's land record quality will exceed even the government's own portals in coverage and accuracy.

2. Case File Accumulation

Every dispute processed through the AI Vakil becomes a training example. After 100,000 case files, the AI becomes dramatically better at Indian land law than any general-purpose LLM. This data does not exist anywhere else.

3. Lawyer Network Lock-In

Lawyers who build their practice on Bhumi (client base, reviews, case history) will not switch to a competitor. This is the same dynamic that made Practo defensible in healthcare.

4. Trust From Social Justice Actions

Every successful alert — a slum saved, an idle land parcel turned into housing — becomes a case study and a press story. This kind of earned trust cannot be bought with marketing spend. It is built one real outcome at a time.

BHUMI — Built for the 66 million Indians who deserve land justice.